

Learning from Inaction

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Motivation

Actions and the absence of actions (*inaction*) can both be informative.

	Action	Inaction
Applications	Receive a response	Receive no response
Trading	A seller reports quality	A seller remains silent
Job market	SUFE hires a behavioral economist	SUFE makes no hire
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- Additional Challenges in Learning from Inaction

Learning from Action

⇒ Infer from the action.

Learning from Inaction

- * Recognize that inaction is informative.
- * Ask when an action would have occurred.

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- Do people underreact to inaction, even when it is as informative as action?

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- **Behavioral biases in belief formation.**

- Correlation neglect [Enke and Zimmermann, 2019; He and Kučinskas, 2024];
 - Hypothetical-contingency neglect [Esponda and Vespa, 2014];
 - Selection neglect [Esponda and Vespa, 2018; Enke, 2020; Barron et al., 2024].
- ⇒ We study a different *action–inaction learning gap*: individuals underreact to inaction relative to informationally equivalent action.

- **Games with informative inaction.**

- Voluntary disclosure game [Jin et al., 2021; Forsythe et al., 1999].
- ⇒ We isolate learning from inaction itself, separating it from strategic withholding and complex equilibrium reasoning.

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What We Do

- We study three settings in which inaction arises naturally from the information structure or the game structure, rather than from strategic communication:
 - **Individual belief updating:** a state-guessing task;
 - **Non-cooperative game:** a colored-hats game;
 - **Cooperative game:** matching with incomplete information.
- Across these three settings, we ask:
 - Is there an *action-inaction learning gap*?
 - If so, can a simple intervention that makes the inaction salient mitigate the gap?

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Experiment 1: State-Guessing Task

Setup: One-Signal Task

- Two states: Red or Blue, with $\Pr(\text{Red}) = 3/5$.
- The signal structure: $\Pr(\text{red} \mid \text{Red}) = 2/3$, $\Pr(\text{red} \mid \text{Blue}) = 1/3$.

Baseline: {red, blue}

$\Pr(\text{blue} \mid \text{Red}) = 1/3$, $\Pr(\text{blue} \mid \text{Blue}) = 2/3$

Signal observed	red	blue
Posterior on Red	3/4	3/7
Bayesian prediction	Red	Blue

Inaction: {red, $\emptyset$ }

$\Pr(\emptyset \mid \text{Red}) = 1/3$, $\Pr(\emptyset \mid \text{Blue}) = 2/3$

Signal observed	red	$\emptyset$
Posterior on Red	3/4	3/7
Bayesian prediction	Red	Blue
Posterior if $\emptyset$ is ignored	3/4	3/5
Behavioral prediction	Red	Red

- Pivotal signal configuration: $\text{blue}(\emptyset)$

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Behavioral prediction Red Red

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Experiment 1: State-Guessing Task

Setup: Two-Signal Task

- Two states: Red or Blue, with $\Pr(\text{Red}) = 2/5$.
- Same signal structure as in the one-signal task; two independent signals.

Baseline: {red, blue}

Signals observed	<i>red-red</i>	<i>mixed</i>	<i>blue-blue</i>
Posterior on Red	8/11	2/5	1/7
Bayesian prediction	Red	Blue	Blue

Inaction: {red, ∅}

Signals observed	<i>red-red</i>	<i>mixed</i>	<i>∅-∅</i>
Posterior on Red	8/11	2/5	1/7
Bayesian prediction	Red	Blue	Blue
Posterior on Red if ∅ is ignored	8/11	4/7	2/5
Behavioral prediction	Red	Red	Blue

- Pivotal signal configuration: *mixed*.

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Experiment 1: Design: 3 Treatments $\times$ 2 Task Sets

- Can explicitly notifying participants when no signal is received improve learning?

⇒ Between-subject design: three treatments.

Baseline: signals are displayed as red or blue.

red ball 1 time, blue ball 1 time  

Inaction: the blue signal is replaced by no signal.

red ball 1 time 

Revealed: same as Inaction, but no signal is explicitly notified.

red ball 1 time  no ball 1 time

⇒ 5 rounds of the one-signal task and 10 rounds of the two-signal task; random order of the two task sets.

- December 2025; 152 SUFE subjects: Baseline (51), Inaction (50), Revealed (51)

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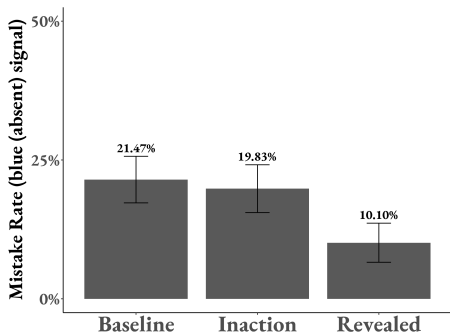
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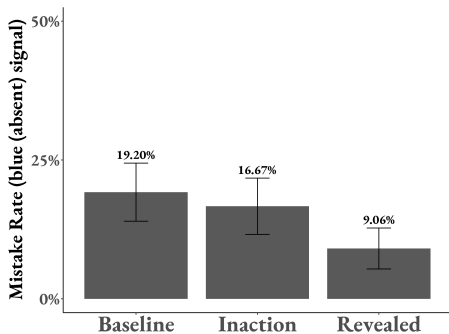
Experiment 1: Results (One-signal Task)

- No action-inaction learning gap in the one-signal task.

For blue (absent) signal, the mistake rate is *Baseline* $\approx$ *Inaction*.



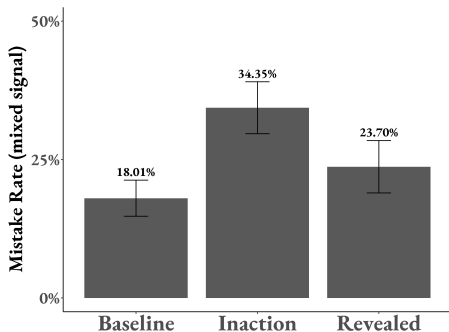
(a) All 5 rounds



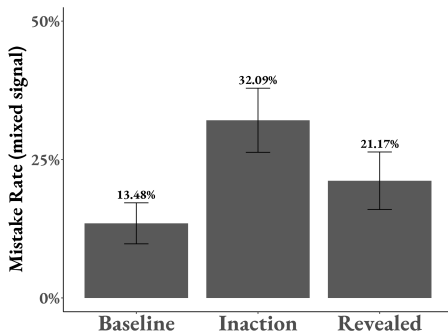
(b) Last 3 rounds

Experiment 1: Results (Two-signal Task)

- For mixed signals, the mistake rate satisfies: *Baseline* $\approx$ *Revealed* $<^{**}$ *Inaction*.



(a) All 10 rounds



(b) Last 5 rounds

Experiment 2: Colored-Hats Game

Setup [Littlewood, 1953]

- Two players each wear a hat, drawn as *Red* or *Blue*.
- A truthful public announcement is made: “At least one hat is Red (or Blue).”
- Each player observes the partner’s hat, but not her own.
- In each period, players guess their own hat color or choose not to guess.

Experiment 2:

Theoretical Prediction

(a) Different-Color Game

	Player 1 (Easy)	Player 2 (Hard)
Public announcement	At least one Red	
Observe partner's hat	Blue	Red
Period 1	Guess Red	$\emptyset$
Period 2		Guess Blue

The hard role learns from the easy role's guess.

(b) Same-Color Game

	Player 1 (Hard)	Player 2 (Hard)
Public announcement	At least one Red	
Observe partner's hat	Red	Red
Period 1	$\emptyset$	$\emptyset$
Period 2	Guess Red	Guess Red

Learning from the absence of the partner's guess.

- For the hard role, the same-color game is more difficult for the hard role because it requires learning from the partner's inaction.
- $\emptyset$ means *Cannot Tell*; representing $\emptyset$ as *Cannot Tell* is theoretically equivalent.

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Experiment 2: Design ($2 * 2$)

- Can representing no guess as an explicit *Cannot Tell* choice improve performance in the same-color game?

⇒ Between-subject variation: two treatments.

- **Baseline**: the action space in each period is $\{\text{Red}, \text{Blue}, \emptyset\}$.
- **Revealed**: the action space in each period is $\{\text{Red}, \text{Blue}, \text{Cannot Tell}\}$.
- Within-subject variation: two types of games across 8 rounds.
In each round, hat colors and the public announcement are randomly drawn, generating *different-color* and *same-color* games.
- April 2026; 114 SUFE subjects: Inaction (58), Revealed (56); Fixed pairing.

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Experiment 2: Procedure and Screenshots

(a) Baseline

Task Round 3

At least one hat is Red 🚒 in your group.

	Correct	Incorrect	Your Choice	Partner's Choice
Period 1	11	-12		Red 🚒
Period 2	7	-8		
Period 3	4	-5		👉
Period 4	2	-3		
Period 5	1	-2		
Period 6	0	0		
...	0	0		

Your partner's hat is: Red 🚒

Current: Period 3

Please choose your hat color:

Red 🚒 Blue 🚒

Double-click to cancel your choice

If you do not choose a color, you may continue choosing in the next period.

Selected: Blue Submit Period 3

(b) Revealed

Task Round 2

At least one hat is Red 🚒 in your group.

	Correct	Incorrect	Cannot Tell	Your Choice	Partner's Choice
Period 1	11	-12	Continue	Cannot Tell	Red 🚒
Period 2	7	-8	Continue		👉
Period 3	4	-5	Continue		
Period 4	2	-3	Continue		
Period 5	1	-2	Continue		
Period 6	0	0	0		
...	0	0	0		

Your partner's hat is: Red 🚒

Current: Period 2

Please choose your hat color:

Red 🚒 Blue 🚒 Cannot Tell

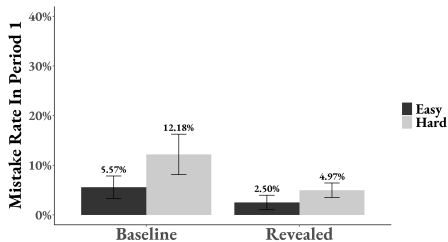
Double-click to cancel your choice

If you choose "Cannot Tell," you may continue choosing in the next period.

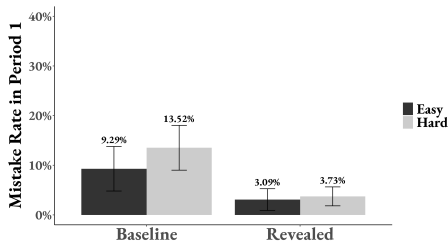
Selected: Blue Submit Period 2

Experiment 2: Results (Individual Mistake Rate in Period 1)

- Period 1 with low mistake rates is not the main source of difficulty.



(a) All 8 Rounds

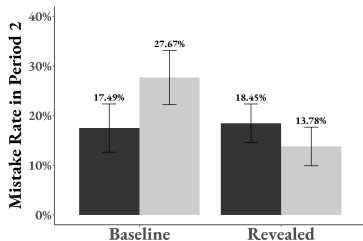


(b) Last 4 Rounds

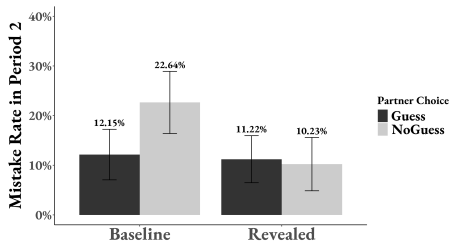
Experiment 2: Results (Hard-role Mistake Rate in Period 2)

Focus on hard-role players who are correct in Period 1 and receive no misleading information from partner.

- Hard-role mistake rates: $LNB >^{**} LB$ under *Baseline*, but $LNB \approx LB$ under *Revealed*.



(a) All 8 Rounds



(b) Last 4 Rounds

Experiment 3: Matching with Incomplete Information

Setup [Liu et al., 2014; Liu, 2020]

- Two workers; two firms
- Two equally likely states: *sunny* or *rainy*; only workers observe the state.
- Payoff matrices are public and state-dependent.

Suppose the true state is *sunny*.

(a) Learning from Blocking (LB)

	Firm 1 (Easy)		Firm 2 (Hard)		State
	1	2	-10	-10	<i>sunny</i>
Worker 1	-1	4	-10	-10	<i>rainy</i>
Worker 2	-10	-10	1	2	<i>sunny</i>
	-10	-10	1	-4	<i>rainy</i>

(b) Learning from No Blocking (LNB)

	Firm 1 (Easy)		Firm 2 (Hard)		
	-1	2	-10	-10	
	1	4	-10	-10	
	-10	-10	1	2	
	-10	-10	1	-4	

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Worker 2	-10	-10	1	2	<i>sunny</i>
	-10	-10	1	-4	<i>rainy</i>

(b) Learning from No Blocking (LNB)

	Firm 1 (Easy)		Firm 2 (Hard)	
Worker 1	-1	2	-10	-10
	1	4	-10	-10
Worker 2	-10	-10	1	2
	-10	-10	1	-4

Experiment 3: Design (2×2)

- Can revealing acceptance and rejection decisions behind matching activity improve performance in LNB?

⇒ Between-subject variation: two treatments.

- *Baseline*: public information includes only changes in matches.
- *Revealed*: public information also includes acceptance and rejection decisions behind attempted matches.
- Within-subject variation: two market types, LB and LNB, across 8 rounds.
 - * 4 LB and 4 LNB rounds; different payoff matrix in each round; random order.
 - * The state is independently drawn in each round.
- April 2026; 124 SJTU subjects: Inaction (64), Revealed (60); Fixed grouping.

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Experiment 3: Decentralized Matching Protocol

(a) Baseline

Timer: 26s Only Lemon and Mango know the weather.

Matching Payoff Table

	Yellow		Green		
Lemon	-2	-5	-8	5	Sunny
	-4	1	6	7	Rainy
Mango	1	1	-7	-6	Sunny
	7	-3	5	-8	Rainy

Current "Temporary Matches" are indicated by the highlighted cells in the Matching Payoff Table.

History "Temporary Matches"

At 118s, Mango was matched with Yellow.
 At 75s, Mango and Yellow were unmatched.
 At 50s, Lemon was matched with Green.
 At 43s, Lemon and Yellow were unmatched.
 At 43s, Mango was matched with Yellow.
 At 35s, Lemon was matched with Yellow.
 At 27s, Mango and Yellow were unmatched.
 At 21s, Lemon and Green were unmatched.
 At 9s, Lemon was matched with Green.

My Role: Green

Weather in this Round: Unknown for me

Sending Invitations

Please choose the Fruit you would like to invite:

Receiving Invitations

Mango sent you an invitation.

Please response within 11s:

My Current "Temporary Match"

Matched with Lemon. Click the button if you would like to dissolve the match:

My Activity History

At 148s, Mango sent me an invitation.
 At 139s, I rejected Mango's invitation.
 At 122s, Mango sent me an invitation.
 At 97s, I accepted Mango's invitation.

(b) Revealed

Timer: 28s Only Lemon and Mango know the weather.

Matching Payoff Table

	Yellow		Green		
Lemon	-2	-8	3	2	Sunny
	-9	-1	-3	1	Rainy
Mango	2	-7	-2	4	Sunny
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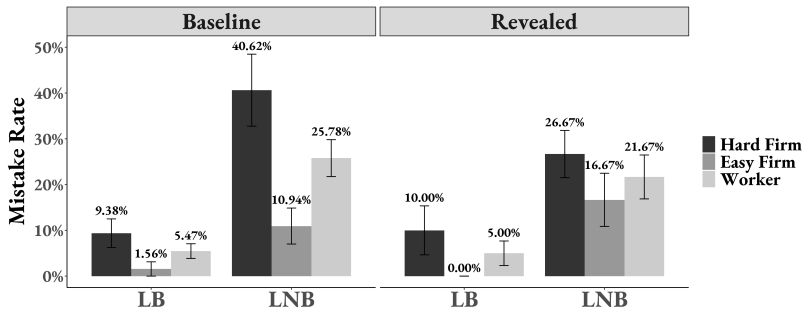
Group Activity History

Filter:

At 78s, Green rejected Lemon's invitation.
 At 55s, Mango accepted Green's invitation, forming an "temporary match".
 At 40s, Lemon accepted Yellow's invitation, forming an "temporary match".
 At 22s, Mango rejected Green's invitation.
 At 9s, Lemon accepted Green's invitation, forming an "temporary match".

Experiment 3: Results (Individual Mistake Rate by Role)

- Hard-firm accounts for the most mistakes.



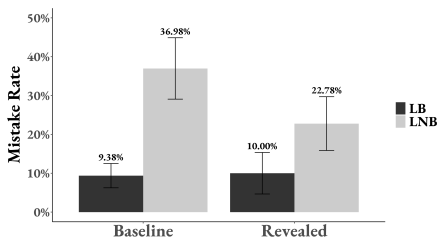
(a) All 8 rounds

- Same pattern for the last four rounds.

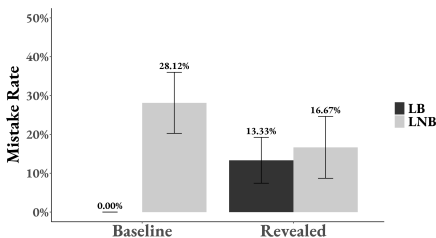
Experiment 3: Results (Hard-firm Mistake Rate)

Conditional on the easy firm acting correctly.

- Hard-firm mistake rates: $LNB >^{**} LB$ under *Baseline*, but $LNB \approx LB$ under *Revealed*.



(a) All 8 rounds



(b) Last 4 rounds

What We Found

- We identify an *action–inaction learning gap*: participants underreact to information conveyed by:
 - the absence of a signal;
 - the absence of another player's choice;
 - market inactivity.
- Simple interventions that make inaction explicit or reveal hidden activity mitigate the gap across all three settings.
 - ⇒ Low salience: inaction may be less likely to trigger belief updating.

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Experiment 1: Screenshots

Two-Signal Task with Mixed Signals:

(a) Baseline

(b) Inaction

(c) Revealed

Round No. 1

Round Setup

40 Red Boxes and 60 Blue Boxes

A "Red Box" contains: 2 red balls, 1 blue ball

A "Blue Box" contains: 1 red ball, 2 blue balls

Computer randomly draws one box in each round.

Computer's Door Opening: 2 Times

Before each door opening, the computer randomly reshuffles the positions of the balls inside the selected box.

The computer has drawn one box



Computer's Door-Opening Feedback

red ball 1 time, blue ball 1 time



Select a Box



Red Box



Blue Box

The computer has drawn one box



Computer's Door-Opening Feedback

red ball 1 time



Select a Box



Red Box



Blue Box

The computer has drawn one box



Computer's Door-Opening Feedback

red ball 1 time, no ball 1 time



Select a Box



Red Box



Blue Box